

QUESTION 1 (10 marks)

- (a) Solve for x : $2x + 7 < 3x + 10$
- (b) What is the size of each interior angle of a regular pentagon?
- (c) The point A is where the line $x + 2y = 8$ intersects the x-axis. Find the coordinates of A.
- (d) Solve the equation : $2 \cos x = \sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$.
- (e) State the largest possible domain for the function $y = \frac{1}{x^2 - 4}$

QUESTION 2 (10 marks)

- (a) Differentiate:
 - (i) $3x^4 - 2x + \frac{1}{x^2} + 1$
 - (ii) $\frac{x}{1+x}$
 - (iii) $(5x + 1)(2x + 3)^3$
- (b) Find the equation of the tangent to the curve $y = 2x^2 + x + 1$ at the point where $x = -1$.

QUESTION 3 (10 marks)

- (a) The function $p(x)$ is defined by the rule: $p(x) = (x - 1)(x^2 - 5)$
 - (i) Find the coordinates of the turning points of $p(x)$ and state whether they are maxima or minima.
 - (ii) Find the real roots of the equation $p(x) = 0$.
 - (iii) Draw a sketch graph of $y = (x - 1)(x^2 - 5)$ for the domain $-3 \leq x \leq 3$. Showing all essential features.
- (b) Solve the quadratic inequality: $x^2 + 7x + 6 > 0$.

QUESTION 4 (10 marks)

- (a) Find $\lim_{h \rightarrow 0} \frac{h^2 + 5h}{h}$
- (b) Find $\frac{ds}{dt}$ if $s = \sqrt{2t^2 - 3t}$
- (c) Find k if $4x^2 - 5x + k = 0$ has equal roots.
- (d) If α and β are the roots of the quadratic equation $2x^2 + 3x + 4 = 0$ find :
 - (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $\alpha^2 + \beta^2$

Solutions to Unit Assessment - Nov '90

Question 1:

2 (a) $2x + 7 < 3x + 10$
 $-x < 3$
 $x > -3$.

2 (b) $\frac{(5-2) \times 180^\circ}{5} = 108^\circ$

2 (c) For A: $y = 0$.

$\therefore x + 8 = 0$
 $x = -8$
 $\therefore A(-8, 0)$

2 (d) $\cos x = \frac{13}{2}$

$\therefore x^\circ = 30^\circ$ and $360^\circ - 30^\circ$
 $= 30^\circ, 330^\circ$



2 (e) All real x except $x = 2$ and $x = -2$.

Question 2:

2 (a) (i) $12x^3 - 2 - 2x^{-3} = 12x^3 - 2 - \frac{2}{x^3}$

2 (ii) $u = x$ $v = x + 1$
 $u' = 1$ $v' = 1$

$y' = \frac{(x+1) \cdot 1 - 1 \cdot x}{(x+1)^2} = \frac{x+1-x}{(x+1)^2} = \frac{1}{(x+1)^2}$

3 (iii) $u = 5x + 1$ $v = (2x + 3)^3$
 $u' = 5$ $v' = 3(2x + 3)^2 \times 2$
 $= 6(2x + 3)^2$

$y' = 5 \times (2x + 3)^3 + 6(2x + 3)^2(5x + 1)$
 $= (2x + 3)^2 [5(2x + 3) + 6(5x + 1)]$
 $= (2x + 3)^2 [10x + 15 + 30x + 6]$
 $= (2x + 3)^2 (40x + 21)$

(b) $y = 2x^2 + x + 1$
 $y' = 4x + 1$ At $x = -1$ $y' = -4 + 1 = -3$

and at $x = -1$ $y = 2(-1)^2 - 1 + 1 = 2$.

$y - 2 = -3(x - -1)$
 $y - 2 = -3x - 3$
 $3x + y + 1 = 0$.

Question 3:

4 (a) (i) $p(x) = (x-1)(x^2-5) = x^3 - 5x - x^2 + 5$
 $= x^3 - x^2 - 5x + 5$

$p'(x) = 3x^2 - 2x - 5$

For turning point $p'(x) = 0$.

ie $3x^2 - 2x - 5 = 0$

$3x \quad -5$
 $x \quad \times \quad 1$

$\therefore (3x-5)(x+1) = 0$

$x = 1$ or $x = -5/3$.

-1.1	-1	-0.9
+	0	-

$\therefore$ Max T.P.

$y = (-1-1)(1-5)$
 $= -2 \times -4$
 $= 8$

$\therefore$ Maxima at $(-1, 8)$

1	5/3	2
-	0	+

$\therefore$ Min T.P.

$y = (\frac{5}{3}-1)(\frac{5}{3}-5)$
 $= -1 \times \frac{10}{3}$
 $= -\frac{10}{3}$

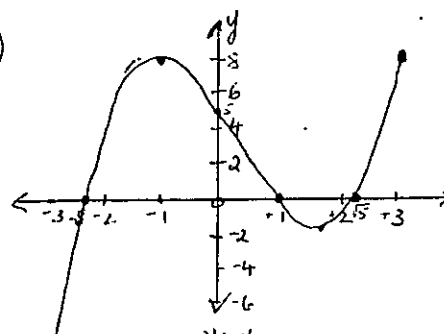
$\therefore$ Minima at $(\frac{5}{3}, -\frac{10}{3})$

2 (ii) $f(x) = 0$

$\therefore (x-1)(x^2-5) = 0$

$\therefore x = 1$ or $\pm\sqrt{5}$.

3 (iii)



correct domain
 x, y intercepts.
 $\frac{1}{2}$ turning pts.

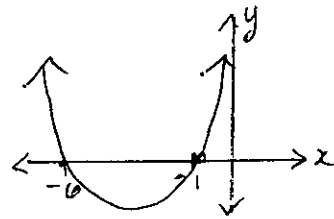
(3)

2 Question 3: (continued)

$$(b) \quad x^2 + 7x + 6 > 0$$

$$(x+1)(x+6) > 0$$

$$\{x: x > -1\} \cup \{x: x < -6\}$$

Question 4:

$$2 (a) \quad \lim_{h \rightarrow 0} \frac{h^2 + 5h}{h} = \lim_{h \rightarrow 0} \frac{h(h+5)}{h} = 5$$

$$2 (b) \quad s = \sqrt{2t^2 - 3t} = (2t^2 - 3t)^{1/2}$$

$$\frac{ds}{dt} = \frac{1}{2} (2t^2 - 3t)^{-1/2} (4t - 3)$$

$$= \frac{4t - 3}{2\sqrt{2t^2 - 3t}}$$

$$2 (c) \quad \text{Equal roots mean } \Delta = 0$$

$$\text{ie } b^2 - 4ac = 0$$

$$\therefore 25 - 4 \times 4 \times k = 0$$

$$25 - 16k = 0$$

$$16k = 25$$

$$k = 25/16 = 1 \frac{9}{16}$$

$$1 (d)(i) \quad \alpha + \beta = -b/a = -3/2$$

$$1 (ii) \quad \alpha\beta = c/a = 4/2 = 2$$

$$2 (iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-3/2)^2 - 2 \times 2$$

$$= \frac{9}{4} - 4$$

$$= -1 \frac{3}{4}$$